

NLP UNIT 6

Applications of NLP

Machine Translation: Rule based techniques, Statistical Machine Translation (SMT), Cross Lingual Translation Sentiment Analysis, Question Answering, Text Entailment, Discourse Processing, Dialog and Conversational Agents, Natural Language Generation

Q. Explain Rule-Based Machine Translation (RBMT) with suitable architecture and example. [8 Marks]

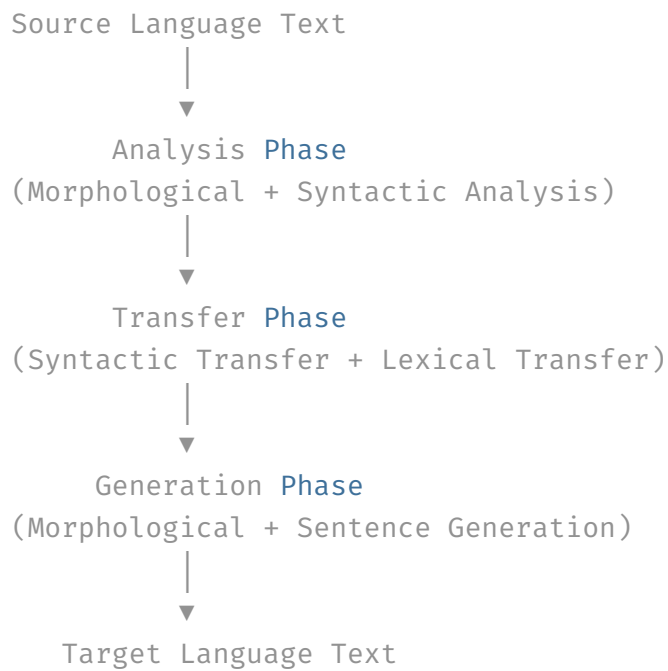
1. Rule-Based Machine Translation (RBMT)

- **Definition** : Rule-Based Machine Translation is a classical machine translation approach that translates text using linguistic knowledge rather than statistical learning.
- **Knowledge Sources** : Uses lexical resources, morphological analyzers, syntactic rules, semantic rules, and bilingual dictionaries to perform translation.

2. Key Principles of RBMT

- **Linguistic Analysis** : Translation is based on explicit language rules and grammatical knowledge.
- **Contrastive Knowledge** : Differences between source and target languages are encoded as transfer rules.
- **Rule Application** : Parsed source structures are transformed into equivalent target-language structures.
- **Dictionary-Based Mapping** : Words and phrases are translated using bilingual lexicons.
- **Meaning Preservation** : Linguistic and transfer rules ensure that the semantic meaning of the source sentence is retained.

3. Architecture of RBMT



- **Analysis Phase** : Parses the source sentence and identifies grammatical, morphological, and syntactic structures.
- **Transfer Phase** : Applies lexical and syntactic transfer rules to convert the source-language representation into a target-language representation.
- **Generation Phase** : Produces a grammatically correct sentence in the target language.

4. Types of RBMT

- **Direct Translation** : Performs word-to-word translation using a bilingual dictionary.
- **Transfer-Based MT** : Uses linguistic analysis and transfer rules between language pairs.
- **Interlingua-Based MT** : Converts source text into a language-independent representation and then generates the target language.

5. Example of RBMT

Source Sentence :

- Mary did not slap the green witch.

Analysis Result :

- Subject → Mary
- Verb → slap
- Negation → not
- Object → the green witch

Transfer Operations :

- did not → no
- slap → dar una bofetada a
- green witch → bruja verde

Target Sentence (Spanish) :

- Maria no dio una bofetada a la bruja verde.

6. Advantages and Limitations

Advantages :

- Produces grammatically controlled translations.
- Easy to incorporate linguistic expertise and domain knowledge.
- Suitable for domain-specific translation systems.
- Translation process is transparent and interpretable.

Limitations :

- Requires extensive rule creation and maintenance.
 - Different language pairs require separate transfer rules.
 - Difficult to handle all language variations and ambiguities.
 - Development is time-consuming and resource-intensive.
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Q. Explain Statistical Machine Translation (SMT) with suitable architecture and example. [8 Marks]

1. Statistical Machine Translation (SMT)

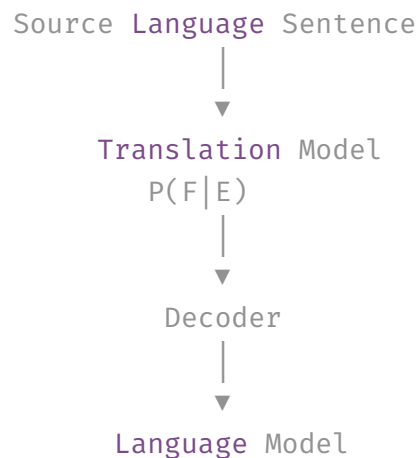
- **Definition** : Statistical Machine Translation (SMT) is a machine translation approach that uses statistical and probabilistic models to generate translations based on patterns learned from bilingual corpora rather than handcrafted linguistic rules.
- **Core Idea** : SMT estimates the probability of a target sentence given a source sentence and selects the most probable translation.
- **Training Data** : Relies on large parallel corpora containing aligned sentences in source and target languages.

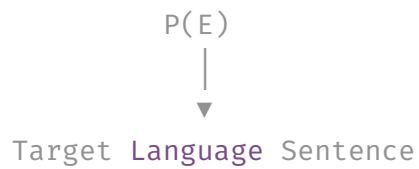
2. Key Principles of SMT

- **Probability-Based Translation** : Translation is treated as a probabilistic decision problem.
- **Learning from Data** : Translation patterns are automatically learned from bilingual text corpora.
- **Phrase-Based Translation** : Sentences are decomposed into phrases or words and translated independently.
- **Word Alignment** : Correspondence between source and target words is learned from parallel sentences.
- **Fluency and Faithfulness** : Best translation maximizes both language fluency and translation accuracy.

3. Architecture of SMT

Auto (SQL) ▾





- **Translation Model** : Computes translation probability between source and target phrases.
- **Decoder** : Searches and selects the most probable translation.
- **Language Model** : Evaluates grammatical correctness and fluency of the target sentence.

4. Noisy Channel Model of SMT

- **Concept** : SMT treats the source sentence as a noisy version of the target sentence and reconstructs the most likely original sentence.
- **Probability Formula** :

$$E = \operatorname{argmax} P(F|E) \times P(E)$$

- $P(F|E)$: Translation Model.
- $P(E)$: Language Model.

5. Example of SMT

Source Sentence (Spanish) :

- Maria no dio una bofetada a la bruja verde

Word/Phrase Alignments :

- Maria → Mary
- no → not
- dio una bofetada → did slap
- bruja verde → green witch

Decoder Output :

- Mary did not slap the green witch.

6. Advantages and Limitations

Advantages :

- Learns translation patterns automatically from data.
- Produces good translations when large parallel corpora are available.
- Supports phrase reordering and probabilistic decision making.

Limitations :

- Requires large bilingual training corpora.
- Performance degrades for low-resource languages.
- May struggle with rare words, ambiguity, and domain-specific terminology.

Q. RBMT vs Statistical Machine Translation (SMT)

Q. Discuss the challenges in Cross-Lingual Translation and provide examples of how it is beneficial in real-world applications. [8 Marks]

1. Cross-Lingual Translation

- **Definition** : Cross-Lingual Translation involves translating text between languages that may not have a direct translation relationship.
- **Pivot Language** : In many cases, an intermediate or pivot language is used to bridge the gap between source and target languages.
- **Objective** : Enables users to access information and communicate across different languages without requiring knowledge of the target language.

2. Translation Ambiguity Challenge

- **Ambiguous Words** : A source word may have multiple meanings, resulting in more than one possible translation.
- **Example** : The word “*Pooja*” may refer to a person's name or the act of worship, causing ambiguity during translation.
- **Impact** : Incorrect interpretation can change the intended meaning of the translated text.

3. Phrase Identification and Structural Differences

- **Word Segmentation** : Some languages, such as Japanese, do not use spaces between words, making phrase identification difficult.
- **Language Variations** : Differences in grammar, sentence structure, tense, and cultural expressions make accurate translation challenging.
- **Meaning Preservation** : Maintaining the original context and semantics across languages is often difficult.

4. Dictionary Coverage and Resource Limitations

- **Dictionary Constraints** : Translation quality depends on the availability of comprehensive linguistic resources.
- **OOV Problems** : New words, technical terms, and domain-specific vocabulary may not exist in dictionaries.
- **Low-Resource Languages** : Limited bilingual corpora and language resources reduce translation accuracy.

5. Pivot Translation Challenges

- **Multi-Step Translation** : Source language is first translated into a pivot language and then into the target language.
- **Error Propagation** : Mistakes introduced in the first translation stage can be amplified in the final output.
- **Semantic Loss** : Original meaning may not be preserved accurately across multiple translation steps.

6. Benefits and Real-World Applications

- **Scientific Research** : Researchers can access articles and publications written in foreign languages.

- **Global News Monitoring** : Organizations can analyze news sources from different countries and languages.
 - **Multilingual Customer Service** : Businesses can provide support to customers speaking different languages.
 - **Patent Analysis** : Companies can examine patents published in multiple languages.
 - **Example** : A French researcher searching for climate-change papers can submit a query in French, retrieve English documents, and obtain translated results in French.
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Q. Write a note on Sentiment Analysis. [10 Marks]

1. Sentiment Analysis

- **Definition** : Sentiment Analysis, also known as Opinion Mining, is the process of determining the sentiment or emotional tone expressed in a piece of text.
- **Objective** : It automatically classifies text as **positive, negative, or neutral** based on the opinions conveyed by words and phrases.
- **Purpose** : Helps organizations understand public opinion, customer attitudes, and user feedback from textual data.

2. Working of Sentiment Analysis

- **Text Collection** : Data is collected from reviews, social media posts, surveys, blogs, and news articles.
- **Preprocessing** : Text is cleaned using tokenization, stop-word removal, and normalization.
- **Feature Extraction** : Important words and phrases are identified using techniques such as Bag-of-Words and TF-IDF.
- **Classification** : The sentiment of the text is determined as positive, negative, or neutral.

3. Techniques Used in Sentiment Analysis

- **Rule-Based Approach** : Uses sentiment lexicons, dictionaries, and manually defined linguistic rules.
- **Machine Learning Approach** : Uses supervised learning and classification algorithms trained on labeled datasets.
- **Deep Learning Approach** : Employs neural networks and pre-trained language models such as BERT for contextual sentiment understanding.
- **Lexicon-Based Method** : Identifies sentiment-bearing words using sentiment dictionaries and lexicons.

4. Types of Sentiment Analysis

- **Document-Level Analysis** : Determines the overall sentiment of an entire document.
- **Sentence-Level Analysis** : Identifies sentiment expressed in individual sentences.
- **Aspect-Based Sentiment Analysis** : Detects sentiment associated with specific product or service features.
 - **Example** : A mobile phone review may express positive sentiment about the camera but negative sentiment about battery life.

5. Applications of Sentiment Analysis

- **Brand Monitoring** : Tracks customer opinions about products and services.
- **Customer Feedback Analysis** : Identifies strengths and weaknesses from reviews.
- **Market Research** : Analyzes consumer preferences and market trends.
- **Reputation Management** : Monitors public perception of organizations.
- **Social Media Monitoring** : Studies public opinion and sentiment trends.
- **Political and Healthcare Analysis** : Assesses public reactions to policies, events, and healthcare services.

6. Challenges in Sentiment Analysis

- **Sarcasm and Irony** : Sentiment may differ from the literal meaning of words.
 - **Context Dependency** : The same word may express different sentiments in different contexts.
 - **Language Nuances** : Handling slang, idioms, abbreviations, and multilingual text is difficult.
 - **Domain Dependency** : Words may carry different sentiment orientations across domains.
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Q. Explain the architecture of a Question Answering (QA) System with neat diagram. [9 Marks]

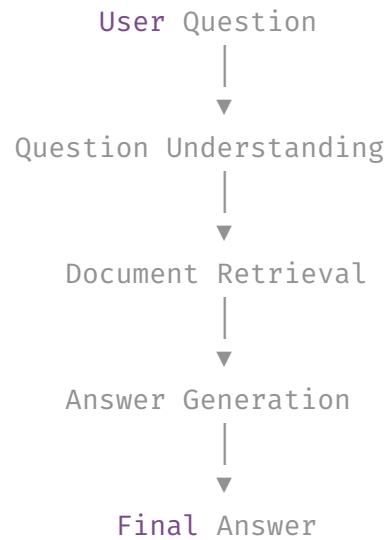
1. Question Answering (QA) System

- **Definition** : Question Answering (QA) is the task of automatically generating accurate answers to questions expressed in natural language.

- **Purpose** : QA systems process user questions, search relevant information sources, and provide concise and precise answers.
- **Applications** : Search engines, virtual assistants, chatbots, customer support systems, and knowledge-based systems.

2. Architecture of Question Answering System

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- **Input** : User submits a natural language question.
- **Processing Flow** : The system analyzes the question, retrieves relevant information, and generates the answer.

3. Question Understanding Stage

- **Function** : Analyzes the user's question to determine its intent and information requirements.
- **Question Type Identification** : Determines whether the question is factual, descriptive, definitional, or open-ended.
- **Keyword Extraction** : Extracts important keywords and entities from the query.
- **Relationship Detection** : Identifies relationships among entities and concepts present in the question.

4. Document Retrieval Stage

- **Function** : Retrieves documents or passages likely to contain the required answer.
- **Search Process** : Uses extracted keywords and entities to search a knowledge base, database, or document collection.
- **Ranking** : Retrieved documents are ranked according to their relevance to the query.
- **Selection** : Most relevant documents or passages are forwarded for answer extraction.

5. Answer Generation Stage

- **Function** : Extracts and generates the most appropriate answer from retrieved documents.
- **Answer Extraction** : Identifies the sentence or passage most likely to contain the answer.
- **NLP Processing** : Uses information retrieval and NLP techniques to pinpoint the exact answer.
- **Response Formatting** : Converts the extracted information into a natural language response suitable for the user.

6. Example of QA System

- **Question** : "Who invented the telephone?"
- **Question Understanding** : Identifies "Who" type question and extracts keyword *telephone*.
- **Document Retrieval** : Searches documents related to the invention of the telephone.
- **Answer Generation** : Extracts the answer *Alexander Graham Bell* and presents it to the user.

7. Advantages of QA Systems

- **Natural Interaction** : Allows users to ask questions in natural language.
 - **Efficient Information Access** : Quickly retrieves relevant information from large document collections.
 - **Accurate Responses** : Provides precise answers instead of returning entire documents.
 - **Wide Applications** : Useful in education, healthcare, search engines, and customer support systems.
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Q. Describe the NLP application: Text Entailment. [10 Marks]

1. Text Entailment

- **Definition** : Text Entailment, also known as Recognizing Textual Entailment (RTE), is the task of determining whether one piece of text logically follows from another.
- **Basic Idea** : It identifies the relationship between two text segments called **Premise** and **Hypothesis** and determines whether the hypothesis can be inferred from the premise.
- **Importance** : It is a fundamental problem in natural language understanding and reasoning.

2. Components of Text Entailment

- **Premise** : Original statement or source text from which inference is made.
- **Hypothesis** : Statement whose validity is checked against the premise.
- **Inference Mechanism** : Determines whether the hypothesis logically follows from the premise.

3. Types of Entailment Relationships

- **Entailment** : The hypothesis logically follows from the premise.
- **Contradiction** : The hypothesis contradicts the information present in the premise.
- **Neutral** : There is insufficient information to determine whether the hypothesis is true or false.

4. Working of Text Entailment System

- **Text Analysis** : Premise and hypothesis are preprocessed and linguistically analyzed.
- **Feature Extraction** : Lexical, syntactic, and semantic features are extracted.
- **Relationship Identification** : Similarities, differences, negations, and semantic relations are examined.
- **Classification** : The pair is classified as Entailment, Contradiction, or Neutral.

5. Approaches Used

- **Rule-Based Methods** : Use linguistic and logical inference rules.

- **Statistical Models** : Learn entailment patterns from annotated datasets.
- **Neural Network Models** : Use deep learning architectures such as RNNs and Transformers for inference.

6. Example

- **Premise** : "A man is riding a bicycle on the road."
- **Hypothesis 1** : "A person is riding a bicycle." → **Entailment**
- **Hypothesis 2** : "A man is driving a car." → **Contradiction**
- **Hypothesis 3** : "The man is going to work." → **Neutral**

7. Applications of Text Entailment

- **Question Answering** : Verifies whether retrieved information answers the question correctly.
- **Information Retrieval** : Improves relevance assessment of retrieved documents.
- **Text Summarization** : Detects redundant information and preserves important facts.
- **Dialogue Systems** : Helps conversational agents understand and validate user statements.

8. Evaluation and Challenges

- **Evaluation Metrics** : Accuracy, Precision, Recall, and F1-Score are commonly used for evaluation.
- **Challenges** : Handling lexical variations, syntactic differences, negation, ambiguity, and quantification remains difficult.
- **Datasets** : SNLI and MNLI datasets are widely used for training and evaluation of entailment systems.

Q. Discuss the key components of a Conversational Agent, such as chatbots or virtual assistants. How do Natural Language Understanding (NLU) and Natural Language Generation (NLG) play a role in creating effective conversational agents? [8 Marks]

1. Conversational Agents

- **Definition** : Conversational Agents are AI systems designed to engage in natural language conversations with users. They are also known as chatbots, virtual assistants, or conversational interfaces.
- **Types** : Task-oriented agents perform specific tasks, while open-domain agents support general conversations.

2. Key Components of Conversational Agent

- **User Interface** : Accepts user input through text or speech.
- **NLU Module** : Understands user intent and extracts relevant information.
- **Dialogue Manager** : Maintains conversation flow and decides the next action.
- **Knowledge Base/Backend** : Stores domain knowledge and external service data.
- **NLG Module** : Generates meaningful responses in natural language.

3. Architecture

Auto (SQL) ▾

User Input



NLU



Dialogue Manager



Knowledge Base



NLG



4. Role of Natural Language Understanding (NLU)

- **Meaning Extraction** : Understands the intended meaning of user utterances through grammar and context analysis.
- **Intent Detection** : Identifies the user's goal or request.
- **Entity Recognition** : Extracts important entities such as names, dates, and locations.
- **Context Handling** : Helps maintain coherent multi-turn conversations.

5. Role of Natural Language Generation (NLG)

- **Response Generation** : Produces natural language output from system decisions or structured data.
- **Lexical Choice** : Selects appropriate words and phrases.
- **Realization** : Generates grammatically correct and fluent responses.
- **Natural Interaction** : Makes conversations more human-like and understandable.

6. Applications

- **Virtual Assistants** : Siri, Alexa, and Google Assistant.
 - **Customer Support** : Automated query handling and troubleshooting.
 - **E-Commerce** : Product recommendation and order tracking.
 - **Healthcare and Banking** : Appointment scheduling and account assistance.
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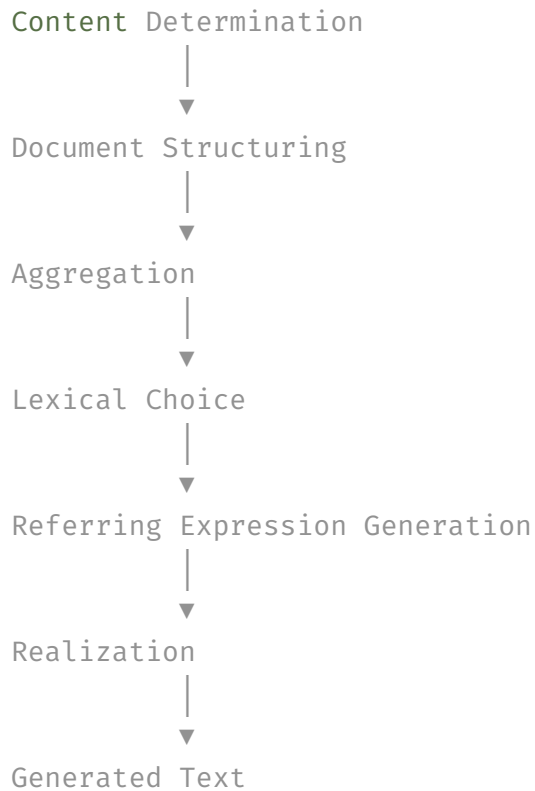
Q. Explain Natural Language Generation (NLG) with reference architecture. [8 Marks]

1. Natural Language Generation (NLG)

- **Definition** : Natural Language Generation is the process of automatically producing human-readable text from structured data or non-linguistic inputs.
- **Purpose** : Converts data into coherent narratives, summaries, descriptions, reports, or responses using computational algorithms and linguistic knowledge.
- **Applications** : Used in report generation, question answering, dialogue systems, machine translation, and text summarization.

2. Reference Architecture of NLG

Auto (CSS) ▾



- **Architecture** : Proposed by Dale and Reiter, consisting of multiple stages that transform data into natural language text.

3. Content Planning Stages

- **Content Determination** : Decides what information should be included in the generated text.

- **Document Structuring** : Organizes information in a logical and meaningful sequence.
- **Aggregation** : Merges related sentences to improve readability and avoid repetition.

4. Linguistic Realization Stages

- **Lexical Choice** : Selects appropriate words and phrases to express concepts.
- **Referring Expression Generation** : Creates references such as names, pronouns, and noun phrases for entities.
- **Realization** : Produces grammatically correct text according to syntax, morphology, and orthography rules.

5. NLG Approaches

- **Template-Based Generation** : Uses predefined templates with placeholders.
- **Rule-Based Generation** : Uses linguistic and grammatical rules.
- **Statistical Generation** : Learns word and phrase probabilities from corpora.
- **Neural NLG** : Uses RNNs, LSTMs, and Transformers to generate fluent text.

6. Applications of NLG

- **Business Reporting** : Automatic generation of financial and analytical reports.
 - **Chatbots and Virtual Assistants** : Generates natural responses during conversations.
 - **Image Captioning** : Produces textual descriptions of images.
 - **Customer Communication** : Creates personalized messages and summaries.
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